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Synopsis
Society Management System

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by

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ABSTRACT

The Society Management System is a web-based application developed to digitize and centralize the day-to-day operations of residential societies. Traditional society management often relies on registers, spreadsheets, paper receipts and separate communication channels, which can make it difficult to maintain records, track payments and dues, manage complaints, monitor visitors and access important information efficiently.

The proposed system provides a centralized platform for managing residents, flats, maintenance bills, payments, dues, complaints, notices, visitors, expenses, staff, documents, notifications and reports. The system supports different categories of users, including Super Admin, Committee, Resident and Security, with access controlled according to their respective roles.

The frontend of the system is developed using React and TypeScript, with Vite as the development and build tool and Tailwind CSS for the user interface. The backend is developed using Python, Django and Django REST Framework (DRF). The frontend communicates with the backend through REST APIs, while JWT-based authentication and role-based access control are used to provide secure access to system functionalities. SQLite is used for data storage.

The proposed system aims to reduce dependency on manual record-keeping, improve data organization, simplify society operations and provide faster access to important information. The system can serve as a scalable digital platform for residential society management and can be further extended with features such as online payment integration, mobile applications, advanced analytics and automated notifications.

C O N T E N T S

- List of Figures i
- List of Tables ii
- Abbreviations and symbols iii

Chapter No.	Topic	Page No.
Chapter 1	Introduction: Background, Problem Statement, Need, Objectives, Scope, Expected Outcomes	
Chapter 2	Literature Review: Survey table, Research Gap, Novelty/Innovation	
Chapter 3	Proposed Methodology: Working Principle, Architecture/Block Diagram, Flowchart, Methodology, Components	
Chapter 4	Project Planning: WBS, Gantt Chart, Roles, Risk Assessment	
Chapter 5	Resources & Feasibility: Technical/Economic/Operational Feasibility, Budget, BOM	
Chapter 6	Expected Results & Impact: Expected outcomes, Industrial relevance, Sustainability, SDG mapping, Future Scope	
Chapter 7	Conclusion	
References	IEEE style; 10–20 references	
Appendix	CAD drawings, calculations, questionnaires, etc. (optional)	

List of Figures

Figure No.	Figure Name	Page No.
1	DFD	25
2	Block Diagram	26
3	Flow Chart	27
4	UML	28
5	GUI 01	29
6	GUI 02	29
7	GUI 03	30

List of Tables

Figure No.	Figure Name	Page No.
1	Survey Table	06 to 09
2	Planning Table	18

Abbreviations and symbols

- **ML**: Machine Learning
- **AI**: Artificial Intelligence
- **TF-IDF**: Term Frequency-Inverse Document Frequency
- **NLP**: Natural Language Processing
- **DFD**: Data Flow Diagram
- **UML**: Unified Modeling Language
- **CSV**: Comma-Separated Values
- **UI**: User Interface
- **PDF**: Portable Document Format
- **HR**: Human Resources
- **ETL**: Extract, Transform, Load

CHAPTER 1: INTRODUCTION

1.1 Background

Introduce the domain and provide the context.

Housing societies involve several day-to-day activities such as resident management, flat records, maintenance billing, payment tracking, complaint management, visitor management, notices, expenses, staff management and document handling.

In many societies, these activities may be managed using physical registers, spreadsheets, paper receipts, messaging applications and separate records. Such approaches can make information difficult to organize, search and monitor.

The proposed **Society Management System** is a web-based application designed to centralize and digitize these operations in a single platform. The system provides different functionalities based on user roles such as Super Admin, Committee, Resident and Security.

The frontend is developed using **React and TypeScript**, while the backend is implemented using **Python, Django and Django REST Framework**. REST APIs are used for communication between the frontend and backend. JWT-based authentication is implemented to provide secure user authentication and role-based access.

1.2 Problem Statement

Traditional society management systems may depend on manual registers, spreadsheets, paper-based receipts and communication platforms such as messaging applications.

This can result in:

- Difficulty in maintaining centralized records.
- Time-consuming searching and updating of information.
- Difficulty in tracking maintenance bills and pending dues.
- Manual management of complaints and notices.
- Difficulty in maintaining visitor records.
- Lack of proper role-based access.
- Duplication of data across different records.
- Difficulty in generating consolidated reports.

Therefore, there is a need for a centralized web-based system that can integrate major society management activities into a single application.

1.3 Need of the Project

The project is needed to provide a centralized platform for managing society-related information and activities.

The system aims to:

- Digitize society records.

- Reduce dependency on paper-based records.
- Centralize resident and flat information.
- Simplify maintenance bill and payment tracking.
- Manage complaints and notices efficiently.
- Maintain visitor and security records.
- Provide role-specific access to users.
- Improve accessibility and organization of society data.
- Generate useful reports from centralized information.

1.4 Objectives

General objective

To develop a centralized web-based Society Management System that digitizes and simplifies common society management operations.

Specific Objectives

1. To develop a web application for managing residents and flat information.
2. To implement maintenance bills, payments and dues management.
3. To provide complaint, notice and notification management.
4. To maintain visitor, staff and security-related records.
5. To manage society expenses, documents and reports.
6. To implement JWT authentication and role-based access control.
7. To provide communication between frontend and backend through REST APIs.

1.5 Scope

The system will cover the following major modules:

Administrative Management

- Resident management
- Flat management
- Staff management
- User management
- Role management

Financial Management

- Maintenance bills
- Payments
- Pending dues
- Expenses
- Financial reports

Society Operations

- Complaints
- Notices
- Notifications
- Visitor records
- Documents

Security

- JWT authentication
- Role-based access control
- Separate access for different user types

User Roles

Role	Main Responsibility
Super Admin	: Overall system and user management
Committee	: Society operations, bills, expenses, complaints and reports
Resident	: View bills, payments, complaints, notices and personal information
Security	: Visitor and security-related management

Limitations

The current system is primarily designed as a web application. Advanced integrations such as dedicated mobile applications, payment gateway integration, biometric devices, smart access-control hardware and automated IoT-based security can be considered for future development.

1.6 Expected Outcomes

The expected outcomes of the project are:

- A functional web-based Society Management System.
- Centralized management of society information.
- Improved organization of resident and flat records.
- Easier tracking of bills, payments and dues.
- Digital complaint and notice management.
- Digital visitor record management.
- Secure role-based access.
- REST API-based frontend-backend communication.
- Generation of useful reports.
- Deployable frontend and backend application.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

1. Web-based residential management systems
2. Society/community management applications
3. Property management systems
4. Role-based access control
5. JWT authentication
6. REST API-based architectures
7. Digital payment and billing management
8. Visitor management systems
9. Complaint management systems
10. Cloud-based management systems

2.2 Literature Review Table

Literature Review Table

Paper	Author	Year	Method	Findings	Research Gap
Society Sync – Digitalize Society Management Systems with Artificial Intelligence Technologies	Mayank Thacker, Lay Shah, Manan Shah	2022	Web/app-based society management with AI, OCR and face recognition	Improves society security, transparency, efficiency and digital record management	Focuses heavily on AI/security features; scope exists for a simpler centralized REST-based system with role-based access
Web App to Manage Residential Societies	Aaryan Chaubey	2023	Web application for maintenance, dues, complaints, notices, communication and security	Automates society activities and improves communication and transparency	Can be extended with modern frontend-backend separation, JWT authentication and detailed role-based permissions
Cloud Based Housing Society Management System	Tirth Shah	2018	Cloud-based web system for property, maintenance payments, notifications, expenses and reports	Centralizes society operations and supports multiple users	Can be enhanced with modern full-stack architecture,

Paper	Author	Year	Method	Findings	Research Gap
					stronger authentication and additional modules such as visitors, staff and documents
Housing Society Management System Using PHP	Vadaliya Harsh et al.	2022	PHP-based web application	Improves complaint handling and allows complaint-status tracking	Mainly complaint-focused; scope for complete multi-module management
Digitalizing Residential Society Management System	Harshal Bolake et al.	2025	Mobile/cloud-based system	Maintenance requests, payments, communication, expense and asset management	Scope for comprehensive web platform with visitors, staff, documents and role-based APIs

2.3 Research Gap

Existing solutions may focus on individual aspects such as resident management, complaint management, visitor management or payment tracking.

The proposed system aims to bring multiple society operations together into a **single centralized platform**, including:
Residents + Flats + Bills + Payments + Dues + Complaints + Visitors + Notices + Expenses + Staff + Documents + Reports

It also incorporates role-based access for different categories of users and separates the frontend and backend through REST APIs.

2.4 Novelty / Innovation

Your novelty can be presented as:

- Centralized management of multiple society operations.
- Multiple role-based dashboards.
- REST API-based architecture.
- JWT authentication.
- Role-based access control.
- Integration of financial, administrative and security modules.
- Digital management of complaints, visitors, notices and documents.
- Web-based access instead of maintaining separate manual records.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System

The proposed system is a full-stack web application consisting of:

Frontend → REST API → Backend → Database

The frontend provides the user interface and dashboards. The backend processes requests, applies business logic and communicates with the database.

3.2 Working Principle

The system works as follows:

1. User opens the Society Management System.
2. User enters login credentials.
3. Backend verifies the credentials.
4. JWT authentication is performed.
5. User role is identified.
6. Appropriate permissions are applied.
7. User accesses permitted modules.
8. React frontend sends HTTP requests to Django REST APIs.
9. Django processes the request.
10. Database is accessed when required.
11. Backend returns JSON response.
12. React displays the requested information.

Example: Residents Module
User
↓
React + TypeScript
↓
GET /api/residents/
↓
Django REST Framework
↓
Database
↓
JSON Response
↓
React
↓
Residents displayed

3.3 System Architecture / Block Diagram

3.4 Flow Chart

START

↓

Open Application

↓

Login

↓

Validate Credentials

↓

Authentication Successful?

|

No —┴—→ Show Error

|

Yes

↓

Identify User Role

↓

Load Role-Based Dashboard

↓

Select Module

↓

Send REST API Request

↓

Django REST Framework

↓

Process Request

↓

Access Database

↓

Return JSON Response

↓

Display Data

↓

Logout

↓

END

3.5 Methodology

The development methodology can be divided into:

Step 1 — Requirement Analysis
Identify society management requirements and user roles.

Step 2 — System Design
Design database structure, frontend components, APIs and access-control architecture.

Step 3 — Frontend Development
Develop the user interface using:

- React
- TypeScript
- Vite
- Tailwind CSS

Step 4 — Backend Development
Develop backend services using:

- Python
- Django
- Django REST Framework

Step 5 — Authentication
Implement:

- JWT authentication
- Login/logout
- Role-based access control

Step 6 — API Development
Create REST APIs for:

- Residents
- Flats
- Bills
- Payments
- Complaints
- Visitors
- Notices
- Expenses
- Staff
- Documents
- Reports

Step 7 — Database Integration
Store and retrieve application data using SQLite.

Step 8 — Testing
Test:

- Authentication
- API endpoints
- CRUD operations
- Role permissions
- Frontend-backend communication
- Form validation

Step 9 — Deployment
Frontend:
React → Netlify
Backend:
Django → Render

3.6 Components / Materials

Table:

Technology	**Purpose**
React	Frontend UI
TypeScript	Type safety
Vite	Development/build tool
Tailwind CSS	UI styling
Axios	API communication
Python	Backend programming
Django	Backend framework
Django REST Framework	REST API development
Simple JWT	Authentication

Technology	Purpose
SQLite	Database
Git	Version control
GitHub	Source-code management
Netlify	Frontend deployment
Render	Backend deployment

CHAPTER 4: PROJECT PLANNING

4.1 Work Breakdown Structure

Society Management System │ ├── Requirement Analysis │ ├── System Design │ ├── UI Design │ ├── Database Design │ └── API Design │ ├── Frontend Development │ ├── Authentication │ ├── Dashboard │ ├── Resident Module │ ├── Billing Module │ ├── Complaint Module │ ├── Visitor Module │ └── Reports │ ├── Backend Development │ ├── Django Setup │ ├── Database │ ├── REST APIs │ ├── JWT Authentication │ └── Role-Based Access │ ├── Integration │ ├── Testing │ ├── Deployment │ └── Documentation

4.2 Gantt Chart

Semester planning

4.3 Roles and Responsibilities

Student	Responsibility
Member 1	React frontend and UI
Member 2	Django backend and REST APIs
Member 3	Database, authentication and authorization
Member 4	Testing, integration and documentation

4.4 Risk Assessment

Risk	Impact	Mitigation
API communication failure	High	API testing and error handling
Database errors	High	Validation and database testing
Authentication issues	High	JWT testing
Unauthorized access	High	Role-based permissions
Deployment failure	Medium	Local testing before deployment
Data inconsistency	Medium	Input validation
Integration problems	Medium	Continuous frontend-backend testing

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

The project is technically feasible because it uses established technologies such as React, TypeScript, Django, Django REST Framework, SQLite and REST APIs.
The required development environment can run on a standard computer capable of running modern web development tools.

5.2 Economic Feasibility

The project is economically feasible because most of the technologies used are open-source.

Resource	**Estimated Cost**
React	₹0
TypeScript	₹0
Vite	₹0
Tailwind CSS	₹0
Python	₹0
Django	₹0
DRF	₹0
SQLite	₹0
Git/GitHub	₹0
Netlify	₹0*
Render	₹0*
Estimated Software Cost	**₹0**

5.3 Operational Feasibility

The system is designed for different categories of users:

- Super Admin
- Committee
- Resident
- Security
-

A web-based interface allows users to access the system through a browser without requiring specialized hardware.

5.4 Budget Estimation

| Item | Quantity | Cost |

5.5 Bill of Materials (BOM)

CHAPTER 6: EXPECTED RESULTS

The proposed system is expected to:

1. Centralize society information.
2. Reduce dependency on manual records.
3. Simplify resident and flat management.
4. Improve maintenance bill and payment tracking.
5. Provide organized complaint management.
6. Improve visitor record management.
7. Provide role-based access to users.
8. Improve accessibility of society information.
9. Provide centralized reports.
10. Provide a scalable architecture for future development.

CHAPTER 7: SUSTAINABILITY & IMPACT

7.1 Industrial Relevance

The system can be useful for:

- Residential housing societies
- Apartment complexes
- Property management companies
- Facility management organizations
- Residential communities

7.2 Societal Impact

The system can improve communication and information accessibility between residents, committee members and security personnel.
It can also reduce the effort required for maintaining records and tracking society activities.

7.3 Environmental Impact

Digitization can reduce dependence on:

- Paper registers
- Printed receipts
- Printed notices
- Physical documentation

This can contribute to reduced paper consumption.

7.4 SDG Mapping

For this software project, the most relevant SDGs can be:

SDG 9 — Industry, Innovation and Infrastructure
The project uses digital technology to improve management infrastructure.

SDG 11 — Sustainable Cities and Communities
The system supports organized management of residential communities.

SDG 12 — Responsible Consumption and Production
Digital records can reduce paper-based documentation.

CHAPTER 8: CONCLUSION

The proposed **Web-Based Society Management System** aims to provide a centralized platform for managing various residential society operations. The system integrates resident and flat management, maintenance bills, payments, complaints, visitors, notices, expenses, staff, documents and reports into a single web application.
The use of **React and TypeScript** provides a structured and interactive frontend, while **Django and Django REST Framework** provide the backend and REST API architecture. JWT authentication and role-based access control help provide appropriate access to Super Admin, Committee, Resident and Security users.
The proposed system is expected to reduce manual record management, improve accessibility of information, simplify society operations and provide a scalable foundation for future enhancements.

REFERENCES

Use IEEE format.

Minimum

- 15–20 research papers
- Books
- Patents
- Standards
- Websites (only if necessary)

APPENDICES (Optional)

- CAD Drawings
- Circuit Diagrams
- Survey Questionnaire
- Data Collection Forms
- Project Canvas
- Ethics Approval (if applicable)